

Hongmin An

(814) 780-1623 | hpa5199@psu.edu | [GitHub](#) | [LeetCode](#)

EDUCATION

The Pennsylvania State University
College of Engineering
Bachelor's in Computer Science

University Park, PA
May 2027
Dean's List (2/4): Fall 2024, Spring 2025

WORK EXPERIENCE

Dispicare

Fabless Display Driver IC Company Solution Team Intern

Seoul, South Korea

June 2025 – August 2025

- Designed and developed an EM dimming algorithm to reduce display panel flicker during pulse-width modulation (PWM), achieving stable luminance across varying brightness levels for a smoother visual experience for end users.
- Built a UART Serial Communication Shell with I²C brightness sensor integration, enabling real-time visualization of brightness levels via dynamic graphs and automated data tables on spreadsheets to improve sensor testing efficiency.
- Collaborated with the application/solutions team to integrate software, hardware, and compression algorithms, gaining hands-on exposure to multiple IPs in a Display Driver IC and reinforcing cross-disciplinary problem-solving skills.

TECHNICAL PROJECTS

EM Dimming Algorithm

Python/Algorithm/Sequences

June 2025 – August 2025

- Reduced display flicker during EM dimming through logic-friendly non-recursive pulse-sequence mapping, including flicker evaluation and Excel-based pulse visualization. [<https://github.com/HongminAn03/EM-Pulse-Algorithm>]

UART Brightness Sensor Serial Shell

Python/C++/Serial Communication

June 2025 – August 2025

- Built a command shell to interface with a UART-connected brightness sensor, enabling graphs with Matplotlib and automated Excel logging for calibration and analysis. [<https://github.com/HongminAn03/UART-Brightness-Sensor>]

Course Scheduler Program

Java/JDBC/GUI

August 2024 – December 2024

- Implemented a scheduling tool with GUI, database integration, and conflict detection logic, providing students with an intuitive interface to manage their academic schedules. [<https://github.com/HongminAn03/CourseScheduler>]

LEADERSHIP

Korean Band Association (KBA)

Founder/President/Lead Guitarist

University Park, PA

January 2025 – Present

- Founded Penn State's first Korean Music Organization, leading 15+ members in performances, enhancing teamwork.

The Bridge Initiative

Volunteer Organization, HR Department Head

University Park, PA

January 2025 – Present

- Led HR team in The Bridge Initiative, coordinating member relations for 30+ volunteers to strengthen engagement.

MILITARY EXPERIENCE

Republic of Korea, Army

Coastal Surveillance Equipment Operator

Ulsan, South Korea

November 2022 – May 2024

- Monitored South Korea's coastal borders using a Thermal Observation Device (TOD) to detect and prevent unauthorized entry and illegal activities, contributing to national security efforts and ensuring national safety.
- Developed a target tracking program that calculates estimated time of arrival using mathematical formulas based on grid coordinates and vessel speed in knots (kts), improving surveillance efficiency and shortening response times.

HONORS/SKILLS/INTERESTS

Languages: Fluent in English, Korean, German **Technology:** Python, JavaScript **Interest:** Game development, Guitar